PROJECT DESIGN PHASE

Solution Architecture – Medical Diagnosis Center

Date	26-09-2026
Team ID	
Project Name	Medical Diagnosis Center by Service Portal (ServiceNow)
Maximum Marks	4 Marks

1. Solution Architecture

The Medical Diagnosis Center by Service Portal is a ServiceNow-based application that connects patients, reception staff, laboratory technicians, and administrators through one portal. It supports patient registration, diagnostic test browsing, appointment booking, sample processing, and viewing or downloading reports.

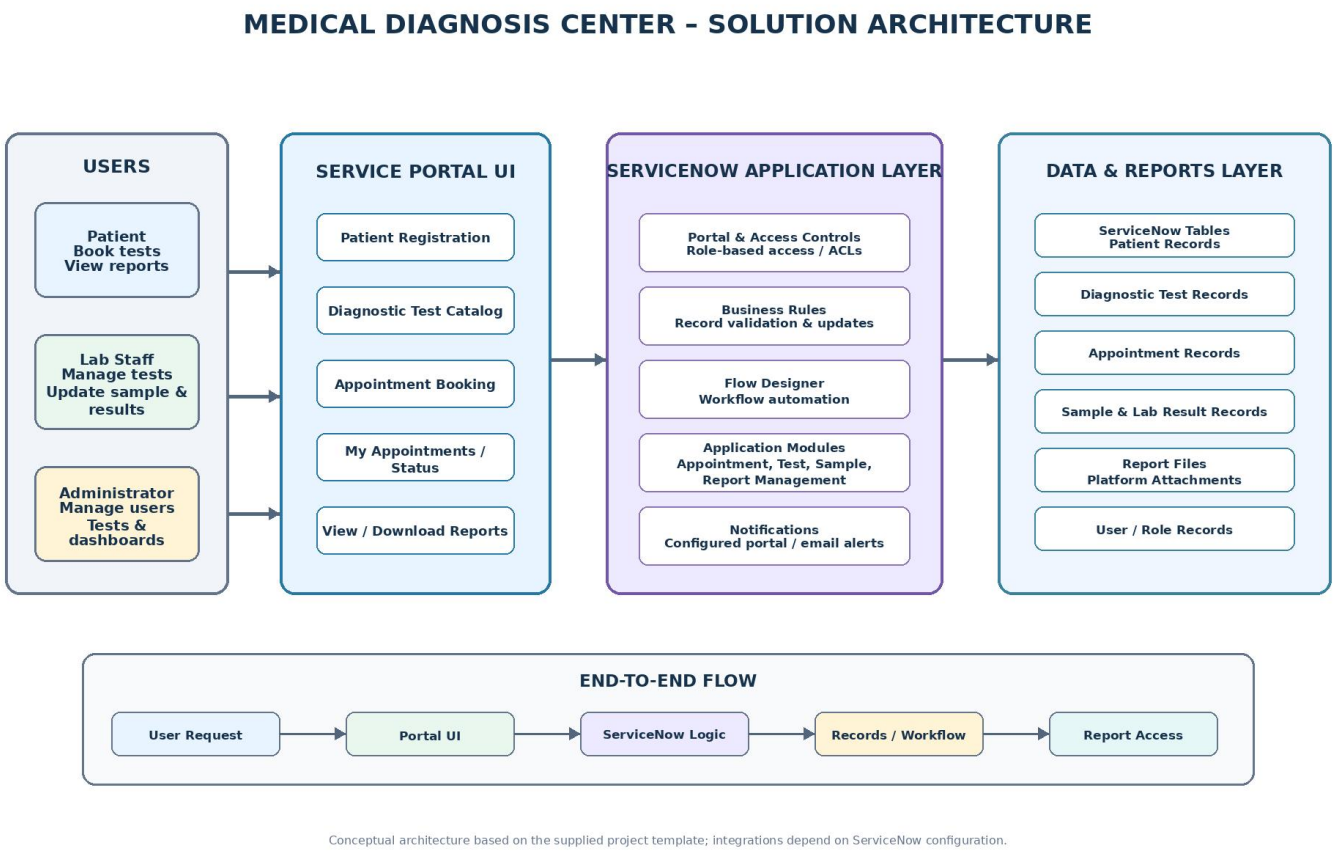


Figure 1. Solution architecture showing users, Service Portal UI, ServiceNow application layer, and data/report storage.

2. Architecture Components and Flow

Component / Actor	Responsibilities / Activities	Technology / Layer
Patient / Reception Staff	Register patients, browse diagnostic tests, book tests and appointments, submit requests, and view booking details.	ServiceNow Service Portal UI
Service Portal UI	Provides browser-based forms and pages for registration, test selection, appointment booking, status tracking, and report access.	HTML, CSS, Client Scripts

ServiceNow Application Layer	Applies portal and access controls, validates records, manages business logic, and coordinates application functions.	ServiceNow platform, Business Rules, Flow Designer
Laboratory Technician	Updates sample collection and diagnostic test status, enters lab results, and makes completed reports available to authorized users.	Lab workflow and application records
Data / Records Layer	Stores patient, diagnostic test, appointment, sample, lab result, and user records in ServiceNow tables.	ServiceNow tables / records
Report Storage	Manages diagnostic report files through platform attachment storage, with access restricted to authorized users.	ServiceNow platform attachments
Administrator	Manages users, tests, appointments, records, and operational dashboards.	ServiceNow administration and role-based access

3. Main Solution Functions

- Patient registration and secure role-based access.
- Diagnostic test catalog and appointment booking.
- Sample collection and laboratory status updates.
- Lab result entry and report viewing or downloading.
- Administrator management of users, tests, and records.

4. System Data Flow

Users interact with the Service Portal to submit or view requests. The ServiceNow application layer processes requests using configured access controls, business rules, and workflows. Patient, test, appointment, sample, and lab-result information is stored in ServiceNow tables. Authorized patients and staff can access reports through the portal, while report files are managed using platform attachments.

Flow: Users → Service Portal → ServiceNow logic and records → Report access